

October 08, 2018

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Fall Biosolids
Pace Project No.: 4617192

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on September 05, 2018. The results relate only to the samples included in this report. Results reported herein conform to the most current, applicable TNI/NELAC standards and the laboratory's Quality Assurance Manual, where applicable, unless otherwise noted in the body of the report.

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Kristin Cothron
kristin.cothron@pacelabs.com
269-381-9666
Project Manager

Enclosures

cc: Accounts Payable, City of Cadillac



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Fall Biosolids

Pace Project No.: 4617192

Grand Rapids Certification ID's

5560 Corporate Exchange Ct SE, Grand Rapids, MI 49512

Minnesota Department of Health, Certificate #1385941

Arkansas Department of Environmental Quality, Certificate
#18-046-0

Georgia Environmental Protection Division, Stipulation

Illinois Environmental Protection Agency, Certificate

#004325

Michigan Department of Environmental Quality, Laboratory

#0034

New York State Department of Health, Serial #57971 and
57972

North Carolina Division of Water Resources, Certificate
#659

Virginia Department of General Services, Certificate #9780

Wisconsin Department of Natural Resources, Laboratory
#999472650

U.S. Department of Agriculture Permit to Receive Soil,
Permit #P330-17-00278

REPORT OF LABORATORY ANALYSIS

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SAMPLE SUMMARY

Project: Fall Biosolids

Pace Project No.: 4617192

Lab ID	Sample ID	Matrix	Date Collected	Date Received
4617192001	Digester #4	Solid	09/04/18 13:45	09/05/18 09:50
4617192002	Confirmation Re-analysis	Solid	09/04/18 13:45	09/05/18 09:50

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SAMPLE ANALYTE COUNT

Project: Fall Biosolids

Pace Project No.: 4617192

Lab ID	Sample ID	Method	Analysts	Analytes Reported
4617192001	Digester #4	EPA 9056A	JHB	3
		EPA 6010C	KLV	15
		EPA 7471B	KLV	1
		SM 2540 G-11/3550	NS1	1
		SM 2540G	NS1	1
		SM 2540G	NS1	1
		EPA 9045D	NS1	1
		EPA 351.2	EB1	1
		SM 4500-NH3 G-11	EB1	1
		SM 4500-P E-11	AVL	1
		EPA 6010C	KLV	1
4617192002	Confirmation Re-analysis	SM 2540 G-11/3550	NS1	1

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ANALYTICAL RESULTS

Project: Fall Biosolids
Pace Project No.: 4617192

Sample: Digester #4 **Lab ID: 4617192001** Collected: 09/04/18 13:45 Received: 09/05/18 09:50 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
9056 IC Anions Analytical Method: EPA 9056A Preparation Method: EPA 9056A								
Chloride	4990	mg/kg	597	1	09/10/18 11:21	09/11/18 02:04	16887-00-6	
Sulfate	448	mg/kg	299	1	09/10/18 11:21	09/11/18 02:04	14808-79-8	
Nitrate as N	<29.9	mg/kg	29.9	1	09/10/18 11:21	09/11/18 02:04	14797-55-8	
6010C MET ICP Analytical Method: EPA 6010C Preparation Method: EPA 3050B								
Aluminum	4460	mg/kg	61.1	1	09/07/18 06:37	09/11/18 09:37	7429-90-5	
Arsenic	61.4	mg/kg	61.1	1	09/07/18 06:37	09/11/18 09:37	7440-38-2	
Cadmium	<6.1	mg/kg	6.1	1	09/07/18 06:37	09/11/18 09:37	7440-43-9	
Calcium	24400	mg/kg	306	1	09/07/18 06:37	09/11/18 09:37	7440-70-2	
Chromium	119	mg/kg	30.6	1	09/07/18 06:37	09/11/18 09:37	7440-47-3	
Copper	897	mg/kg	6.1	1	09/07/18 06:37	09/11/18 09:37	7440-50-8	
Lead	32.0	mg/kg	30.6	1	09/07/18 06:37	09/11/18 09:37	7439-92-1	
Magnesium	4440	mg/kg	306	1	09/07/18 06:37	09/11/18 09:37	7439-95-4	
Molybdenum	<61.1	mg/kg	61.1	1	09/07/18 06:37	09/11/18 09:37	7439-98-7	
Nickel	29.1	mg/kg	6.1	1	09/07/18 06:37	09/11/18 09:37	7440-02-0	
Potassium	2890	mg/kg	306	1	09/07/18 06:37	09/11/18 09:37	7440-09-7	
Selenium	<61.1	mg/kg	61.1	1	09/07/18 06:37	09/11/18 09:37	7782-49-2	
Silver	<6.1	mg/kg	6.1	1	09/07/18 06:37	09/11/18 09:37	7440-22-4	
Sodium	4460	mg/kg	306	1	09/07/18 06:37	09/11/18 09:37	7440-23-5	
Zinc	1170	mg/kg	30.6	1	09/07/18 06:37	09/11/18 09:37	7440-66-6	
7471 Mercury Analytical Method: EPA 7471B Preparation Method: EPA 7471B								
Mercury	0.51	mg/kg	0.076	1	09/27/18 09:23	09/28/18 10:11	7439-97-6	
Percent Moisture Analytical Method: SM 2540 G-11/3550								
Percent Moisture	96.7	%	0.10	1		09/07/18 16:47		
2540G Total Percent Solids Analytical Method: SM 2540G								
Total Solids	3.3	%	0.10	1		09/07/18 16:47		D6,N2
2540G Total Volatile Solids Analytical Method: SM 2540G								
Total Volatile Solids	56.3	%	0.10	1		09/07/18 16:47		
9045 pH Soil Analytical Method: EPA 9045D								
pH at 25 Degrees C	7.8	Std. Units	1.0	1		09/06/18 14:51		
351.2 Total Kjeldahl Nitrogen Analytical Method: EPA 351.2 Preparation Method: EPA 351.2								
Nitrogen, Kjeldahl, Total	57000	mg/kg	10900	5	09/06/18 11:30	09/06/18 18:58	7727-37-9	
4500 Ammonia Soil, Distilled Analytical Method: SM 4500-NH3 G-11 Preparation Method: SM 4500-NH3 B-11								
Nitrogen, Ammonia	22300	mg/kg	306	10	09/17/18 16:41	09/19/18 14:36	7664-41-7	
4500PE Total Phosphorus Analytical Method: SM 4500-P E-11 Preparation Method: SM 4500-P B-11								
Phosphorus	48200	mg/kg	3040	10	09/13/18 10:34	09/14/18 11:15	7723-14-0	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Fall Biosolids

Pace Project No.: 4617192

Sample: Confirmation Re-analysis **Lab ID: 4617192002** Collected: 09/04/18 13:45 Received: 09/05/18 09:50 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	PQL	DF	Prepared	Analyzed	CAS No.	Qual
6010C MET ICP								
Analytical Method: EPA 6010C Preparation Method: EPA 3050B								
Arsenic	45.9J	mg/kg	61.0	1	09/28/18 06:40	10/01/18 12:48	7440-38-2	
Percent Moisture								
Analytical Method: SM 2540 G-11/3550								
Percent Moisture	96.7	%	0.10	1		10/05/18 13:51		H1

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch: 32789

Analysis Method: EPA 9056A

QC Batch Method: EPA 9056A

Analysis Description: 9056 IC Anions

Associated Lab Samples: 4617192001

METHOD BLANK: 132214

Matrix: Solid

Associated Lab Samples: 4617192001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<19.0	19.0	09/10/18 20:46	
Nitrate as N	mg/kg	<0.95	0.95	09/10/18 20:46	
Sulfate	mg/kg	<9.5	9.5	09/10/18 20:46	

LABORATORY CONTROL SAMPLE: 132215

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	493	474	96	80-120	
Nitrate as N	mg/kg	29.6	28.2	95	70-130	
Sulfate	mg/kg	493	464	94	70-130	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 132217 132218

Parameter	Units	4616988001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	Max RPD	Qual
Chloride	mg/kg	31.4	519	497	510	493	92	93	80-120	3	20
Nitrate as N	mg/kg	<1.0	31.1	29.8	29.7	28.5	94	94	70-130	4	20
Sulfate	mg/kg	57.9	519	497	554	531	96	95	70-130	4	20

SAMPLE DUPLICATE: 132216

Parameter	Units	4616988001 Result	Dup Result	RPD	Max RPD	Qualifiers
Chloride	mg/kg	31.4	32.0	2	20	
Nitrate as N	mg/kg	<1.0	<1.0		20	
Sulfate	mg/kg	57.9	58.8	1	20	

SAMPLE DUPLICATE: 132483

Parameter	Units	4617343001 Result	Dup Result	RPD	Max RPD	Qualifiers
Chloride	mg/kg	1430	1370	4	20	
Nitrate as N	mg/kg	64.0	64.2	0	20	
Sulfate	mg/kg	1060	1060	0	20	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch: 34269

Analysis Method: EPA 7471B

QC Batch Method: EPA 7471B

Analysis Description: 7471 Mercury

Associated Lab Samples: 4617192001

METHOD BLANK: 138401

Matrix: Solid

Associated Lab Samples: 4617192001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.047	0.047	09/28/18 08:40	

LABORATORY CONTROL SAMPLE: 138402

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	.31	0.28	92	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 138403

138404

Parameter	Units	4618003001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	<0.060	.38	.39	0.36	0.38	87	90	80-120	6	20	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch:	32525	Analysis Method:	EPA 6010C
QC Batch Method:	EPA 3050B	Analysis Description:	6010 MET
Associated Lab Samples:	4617192001		

METHOD BLANK: 131329 Matrix: Solid

Associated Lab Samples: 4617192001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<9.9	9.9	09/11/18 08:24	
Arsenic	mg/kg	<9.9	9.9	09/11/18 08:24	
Cadmium	mg/kg	<0.99	0.99	09/11/18 08:24	
Calcium	mg/kg	<49.6	49.6	09/11/18 08:24	
Chromium	mg/kg	<5.0	5.0	09/11/18 08:24	
Copper	mg/kg	<0.99	0.99	09/11/18 08:24	
Lead	mg/kg	<5.0	5.0	09/11/18 08:24	
Magnesium	mg/kg	<49.6	49.6	09/11/18 08:24	
Molybdenum	mg/kg	<9.9	9.9	09/11/18 08:24	
Nickel	mg/kg	<0.99	0.99	09/11/18 08:24	
Potassium	mg/kg	<49.6	49.6	09/11/18 08:24	
Selenium	mg/kg	<9.9	9.9	09/11/18 08:24	
Silver	mg/kg	<0.99	0.99	09/11/18 08:24	
Sodium	mg/kg	<49.6	49.6	09/11/18 08:24	
Zinc	mg/kg	<5.0	5.0	09/11/18 08:24	

LABORATORY CONTROL SAMPLE: 131330

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	99.8	95.5	96	80-120	
Arsenic	mg/kg	99.8	99.9	100	80-120	
Cadmium	mg/kg	20	19.9	100	80-120	
Calcium	mg/kg	998	932	93	80-120	
Chromium	mg/kg	20	20.0	100	80-120	
Copper	mg/kg	20	20.1	101	80-120	
Lead	mg/kg	20	19.9	100	80-120	
Magnesium	mg/kg	998	984	99	80-120	
Molybdenum	mg/kg	20	20.8	104	80-120	
Nickel	mg/kg	20	20.0	100	80-120	
Potassium	mg/kg	998	942	94	80-120	
Selenium	mg/kg	99.8	97.0	97	80-120	
Silver	mg/kg	20	19.9	99	80-120	
Sodium	mg/kg	998	973	98	80-120	
Zinc	mg/kg	20	20.0	100	80-120	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 131331				131332								
Parameter	Units	4616826003	MS	MSD	MS	MSD	MS	MSD	% Rec	Max	Qual	
		Result	Spike Conc.	Spike Conc.								Result
Aluminum	mg/kg	20700	96	99.2	24000	21900	3410	1200	75-125	9	20	M6
Arsenic	mg/kg	<99.8	96	99.2	<96.0	<99.2	56	52	75-125		20	M6
Cadmium	mg/kg	140	19.2	19.8	169	143	150	16	75-125	17	20	M6
Calcium	mg/kg	<499	960	992	739	751	71	70	75-125	2	20	M6
Chromium	mg/kg	17100	19.2	19.8	16600	15200	-2790	-9390	75-125	8	20	M6
Copper	mg/kg	709	19.2	19.8	715	660	35	-247	75-125	8	20	M6
Lead	mg/kg	<49.9	19.2	19.8	<48.0	<49.6	8	105	75-125		20	M6
Magnesium	mg/kg	<499	960	992	<480	<496	19	20	75-125		20	M6
Molybdenum	mg/kg	866	19.2	19.8	934	867	357	7	75-125	7	20	M6
Nickel	mg/kg	14800	19.2	19.8	14700	13600	-536	-6000	75-125	8	20	M6
Potassium	mg/kg	90700	960	992	82600	84000	-841	-675	75-125	2	20	M6
Selenium	mg/kg	<99.8	96	99.2	<96.0	<99.2	10	0	75-125		20	M6
Silver	mg/kg	<10	19.2	19.8	20.9	21.5	104	103	75-125	3	20	
Sodium	mg/kg	580	960	992	1400	1550	86	98	75-125	10	20	
Zinc	mg/kg	101000	19.2	19.8	97400	109000	-16800	44300	75-125	12	20	E,M6

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch: 34311

Analysis Method: EPA 6010C

QC Batch Method: EPA 3050B

Analysis Description: 6010 MET

Associated Lab Samples: 4617192002

METHOD BLANK: 138513

Matrix: Solid

Associated Lab Samples: 4617192002

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	1.7U	10	10/01/18 11:58	

LABORATORY CONTROL SAMPLE: 138514

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	92.2	86.0	93	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 138515

138516

Parameter	Units	40176249001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	73.8J	159	160	220J	204J	92	81	75-125		20	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch: 32683

Analysis Method: SM 2540 G-11/3550

QC Batch Method: SM 2540 G-11/3550

Analysis Description: Dry Weight/Percent Moisture

Associated Lab Samples: 4617192001

SAMPLE DUPLICATE: 131851

Parameter	Units	4617068010 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	69.5	67.3	3	20	

SAMPLE DUPLICATE: 131852

Parameter	Units	4617192001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	96.7	96.9	0	20	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch: 35088

Analysis Method: SM 2540 G-11/3550

QC Batch Method: SM 2540 G-11/3550

Analysis Description: Dry Weight/Percent Moisture

Associated Lab Samples: 4617192002

SAMPLE DUPLICATE: 141795

Parameter	Units	4617192002 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	96.7	96.9	0	20	H1

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch: 32667

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Solids

Associated Lab Samples: 4617192001

METHOD BLANK: 131836

Matrix: Solid

Associated Lab Samples: 4617192001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Total Solids	%	<0.10	0.10	09/07/18 16:46	N2

SAMPLE DUPLICATE: 131837

Parameter	Units	4617192001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	3.3	3.1	6	5	D6,N2

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch: 32668

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Volatile Solids

Associated Lab Samples: 4617192001

SAMPLE DUPLICATE: 131838

Parameter	Units	4617192001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	56.3	59.5	5	5	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch: 32538

Analysis Method: EPA 9045D

QC Batch Method: EPA 9045D

Analysis Description: 9045 pH

Associated Lab Samples: 4617192001

LABORATORY CONTROL SAMPLE: 131377

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
pH at 25 Degrees C	Std. Units	6	6.0	101	99-101	

SAMPLE DUPLICATE: 131378

Parameter	Units	4617170001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	7.6	7.7	1	20	

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch: 32546

Analysis Method: EPA 351.2

QC Batch Method: EPA 351.2

Analysis Description: 351.2 TKN

Associated Lab Samples: 4617192001

METHOD BLANK: 131411

Matrix: Solid

Associated Lab Samples: 4617192001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<100	100	09/06/18 18:32	

LABORATORY CONTROL SAMPLE: 131412

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	250	247	99	90-110	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 131413

131414

Parameter	Units	4617170001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	44300	1690	1770	58600	66800	849	1270	90-110	13	20	M6

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch: 33312

Analysis Method: SM 4500-NH3 G-11

QC Batch Method: SM 4500-NH3 B-11

Analysis Description: 4500 Ammonia, Distilled

Associated Lab Samples: 4617192001

METHOD BLANK: 134566

Matrix: Solid

Associated Lab Samples: 4617192001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<1.0	1.0	09/19/18 12:09	

LABORATORY CONTROL SAMPLE: 134567

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	20	19.1	96	82-112	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 134568

134569

Parameter	Units	4617170001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	903	140	136	1270	1230	265	238	49-148	4	20	M1

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QUALITY CONTROL DATA

Project: Fall Biosolids

Pace Project No.: 4617192

QC Batch: 33107

Analysis Method: SM 4500-P E-11

QC Batch Method: SM 4500-P B-11

Analysis Description: 4500PE Soil Phosphorus

Associated Lab Samples: 4617192001

METHOD BLANK: 133598

Matrix: Solid

Associated Lab Samples: 4617192001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Phosphorus	mg/kg	<10.0	10.0	09/14/18 10:38	

LABORATORY CONTROL SAMPLE: 133599

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phosphorus	mg/kg	100	97.7	98	90-110	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 133600

133601

Parameter	Units	4617170001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Phosphorus	mg/kg	4780	625	659	5130	5320	55	81	90-110	4	20	M1

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALIFIERS

Project: Fall Biosolids

Pace Project No.: 4617192

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

D6	The precision between the sample and sample duplicate exceeded laboratory control limits.
E	Analyte concentration exceeded the calibration range. The reported result is estimated.
H1	Analysis conducted outside the EPA method holding time.
M1	Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.
M6	Matrix spike and Matrix spike duplicate recovery not evaluated against control limits due to sample dilution.
N2	The lab does not hold NELAC/TNI accreditation for this parameter.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Fall Biosolids

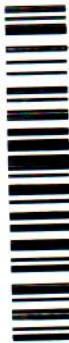
Pace Project No.: 4617192

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
4617192001	Digester #4	EPA 9056A	32789	EPA 9056A	32862
4617192001	Digester #4	EPA 3050B	32525	EPA 6010C	32883
4617192002	Confirmation Re-analysis	EPA 3050B	34311	EPA 6010C	34528
4617192001	Digester #4	EPA 7471B	34269	EPA 7471B	34389
4617192001	Digester #4	SM 2540 G-11/3550	32683		
4617192002	Confirmation Re-analysis	SM 2540 G-11/3550	35088		
4617192001	Digester #4	SM 2540G	32667		
4617192001	Digester #4	SM 2540G	32668		
4617192001	Digester #4	EPA 9045D	32538		
4617192001	Digester #4	EPA 351.2	32546	EPA 351.2	32716
4617192001	Digester #4	SM 4500-NH3 B-11	33312	SM 4500-NH3 G-11	33610
4617192001	Digester #4	SM 4500-P B-11	33107	SM 4500-P E-11	33259

REPORT OF LABORATORY ANALYSIS

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WO#: 4617192



4617192

CHAIN-OF-CUSTODY / Analytical Request Document

The Chain-of-Custody is a LEGAL DOCUMENT. All relevant fields must be completed accurately.

Section C

Project Information:

Company: City of Cadillac
 Address: 200 North Lake St.
 Cadillac, MI 49601
 Email: ctomaszewski@cadillac-mi.net
 Phone: (231) 775-2368
 Requested Due Date:

Invoice Information:

Report To: Cindy Tomaszewski
 Copy To:
 Purchase Order #:
 Project Name: Fall Biosolids
 Project #:
 Attention: Cindy Tomaszewski
 Company Name:
 Address:
 Pace Project Manager: kristin.cothron@pacelabs.com,
 Pace Profile #:

Regulatory Agency

MDAQ

State / Location

MI

ITEM #	MATRIX	CODE	COLLECTED		SAMPLE TYPE (G=GRAB C=COMP)	MATRIX CODE (see valid codes to left)	SAMPLE TEMP AT COLLECTION	# OF CONTAINERS	Unpreserved	H2SO4	HNO3	HCl	NaOH	Na2S2O3	Methanol	Other	Analyses Test Y/N	Metals, Nutrients	Residual Chlorine (Y/N)	Sludge
			START	END																
1	Digester #4		DATE	TIME	DATE	TIME	9/4	1315												
2																				
3																				
4																				
5																				
6																				
7																				
8																				
9																				
10																				
11																				
12																				

ADDITIONAL COMMENTS	RELINQUISHED BY / AFFILIATION	DATE	TIME	ACCEPTED BY / AFFILIATION	DATE	TIME	TEMP in C	Received on	Ice (Y/N)	Custody	Sealed Cooler (Y/N)	Samples Intact (Y/N)
City of Cadillac	City of Cadillac	9/4/18	1400	Ship via UPS	9/5/18	0950						

SAMPLER NAME AND SIGNATURE: TAYSHA LACZKO
 PRINT Name of SAMPLER: TAYSHA LACZKO
 SIGNATURE of SAMPLER: [Signature]
 DATE Signed: 9/4/2018



Sample Conditions Upon Receipt

WO#: 4617192

Date: 9/18/18	Evaluated by: PS	PM: KMC	Due Date: 09/19/18
Client: City of Cecilice	Project Manager:	CLIENT: KZ-CAD	
Sample Receiving Non-Conformance Form Required: YES	NO	Rush Turn Around Time Requested: YES	NO
Page Of	Lab Notified of Rush or Short Holds: YES	NO	

Lab Sample Receipt Checklist:

Samples Received Via:	FEDEX	UPS	CLIENT	PACE COURIER
Custody Seals Present and Intact:	YES	NO	NA	
USDA Regulated Soils:	YES	NO	NA	
Short Holds Present (< 72 Hours):	YES	NO	NA	
Samples Received in Hold:	YES	NO	NA	
Custody Signatures Present:	YES	NO	NA	
Collector Signature Present:	YES	NO	NA	
Samples Received On Ice: Bagged Ice	YES	NO	NA	
Type of Ice: WET BLUE DRY NONE	YES	NO	NA	
Packing Material Used:	YES	NO	NA	
IR Gun #: 202 329 402 Temp should be 0-6°C	Cooler Temp Upon Receipt: 3.7 °C			
Temp Blank Received:	YES	NO	NA	
Trip Blank Received: Type: HCL MeOH TSP OTHER	YES	NO	NA	
Bottles Intact:	YES	NO	NA	
Correct Bottles:	YES	NO	NA	
Sufficient Volume:	YES	NO	NA	
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation Exceptions are VOA, coliform, TOC, O & G, HEM, DRO	YES	NO	NA	pH Strip Lot Number: HC 739245
VOA Headspace Acceptable (<6mm):	YES	NO	NA	
Comments:				